MELVIN MARK MICHAEL

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Available for Professional Attachment: September 2026

Professional Objective

Final-year BSc Computer Science and Data Science student at Raffles University Malaysia seeking an internship at Brooks Automation Malaysia, Pasir Gudang, available from September 2026. Uniquely positioned for a semiconductor automation company — my Final Year Project involved building an AI-powered defect classification system for semiconductor wafer manufacturing, achieving 95.74% accuracy on 811,457 real-world production samples. Experienced in Python, deep learning, computer vision, and manufacturing intelligence systems.

Education

BSc (Hons) Computer Science and Data Science *2023 – 2026 (Expected)*
Raffles University Malaysia | Iskandar Puteri, Johor Bahru

Relevant Coursework:

- Machine Learning and AI: deep learning, CNN, LSTM, model training, evaluation, and deployment for manufacturing applications
- Computer Vision: image classification, convolutional neural networks, defect pattern recognition, feature extraction
- Data Analytics: statistical analysis, pattern recognition, predictive modelling, business reporting
- Software Engineering: SDLC, system documentation, Agile methodologies, requirements analysis

Final Year Project: AI-Powered Wafer Map Defect Classification for Semiconductor Manufacturing

- Designed Hybrid CNN with Squeeze-and-Excitation (SE) attention blocks for automated semiconductor wafer defect recognition on WM811K dataset (811,457 samples, 9 defect classes) — directly relevant to Brooks Automation's precision quality control and manufacturing automation
- Results: 95.74% accuracy, 83.88% Macro F1, 93.09% Macro Recall — outperforming baseline CNN; 3-seed benchmark vs ResNet50 and EfficientNet-B3; deployed as live Streamlit inference dashboard

Technical Projects

Predictive Maintenance System: Industrial Equipment *2026*
Python, PyTorch, LSTM, SHAP, Streamlit | github.com/melvinmark9981/predictive-maintenance-cmapss

- AI-powered manufacturing equipment health monitoring using NASA CMAPSS dataset — predicts Remaining Useful Life (RUL) via LSTM, with real-time sensor trend charts, failure risk alerts, and SHAP feature importance; directly applicable to Brooks Automation's semiconductor equipment monitoring

Malaysia Smart City IoT Dashboard *2026*

Python, Streamlit, Plotly | github.com/melvinmark9981/malaysia-smart-city-dashboard

- Real-time IoT sensor monitoring system with live Z-score anomaly detection, city heatmaps, and zone-level KPI dashboards — demonstrates sensor data processing and manufacturing monitoring capabilities

English-Spanish Neural Machine Translation System *2025*

Python, Whisper ASR, mBART-50, gTTS, Gradio | NLP Module Project

- End-to-end AI speech pipeline achieving BLEU 27.81 to 28.44 via LoRA fine-tuning on mBART-50, a 34% improvement over base model

Work Experience

Founder and Operations Director *2022 – Present*

DWIN CLUB | Johor Bahru, Malaysia

- Managed large-scale operations across 15+ events — project coordination, SOPs, vendor management, digital reporting; demonstrates ability to work in fast-paced environments and deliver results under pressure

Insurance Agent *2023 – Present*

R. Michael Agency, Allianz Malaysia / MSIG | Johor Bahru, Malaysia

- Client lifecycle management, data entry, policy documentation, and reporting using digital portals

Technical Skills

AI and Deep Learning: PyTorch, scikit-learn, CNN, LSTM, Squeeze-and-Excitation, Transformer, Knowledge Distillation, SHAP

Computer Vision: Wafer defect classification, image classification, Grad-CAM visualisation, feature extraction

Manufacturing and IoT: Sensor data processing, real-time monitoring, anomaly detection, predictive maintenance

Data Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Folium, statistical analysis, predictive modelling

Programming Languages: Python, JavaScript, TypeScript, C++, SQL, HTML, CSS, Bash

Software Engineering: SDLC, system documentation, Agile, Git, GitHub, VS Code, REST APIs

Web and Deployment: Streamlit, Gradio, Next.js, React, Supabase, Vercel, Node.js

Key Achievements

- FYP: Hybrid CNN-SE model achieved 95.74% accuracy and 83.88% Macro F1 on WM811K semiconductor wafer dataset — highest performance across all benchmarked architectures (Residual-SE vs ResNet50 vs EfficientNet-B3) in rigorous 3-seed evaluation; deployed as live Streamlit inference dashboard for real-time wafer defect detection
- Predictive Maintenance System (github.com/melvinmark9981/predictive-maintenance-cmapss): end-to-end AI manufacturing equipment monitoring using NASA CMAPSS dataset — LSTM RUL prediction, real-time sensor charts, failure risk alerts, and SHAP explainability; directly applicable to Brooks Automation's semiconductor equipment monitoring
- Malaysia Smart City IoT Dashboard (github.com/melvinmark9981/malaysia-smart-city-dashboard): live IoT platform monitoring 5 Malaysian cities with real-time Z-score anomaly detection, GPS-accurate heatmaps, and KPI dashboards — sensor data processing skills transferable to manufacturing floor monitoring

Languages

- English: Fluent, professional written and spoken
- Bahasa Malaysia: Native
- Tamil: Conversational